

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                           |                    |             |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------|-------------|
| Programme      | B. Sc. Mathematics Honours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                           |                    |             |
| Course Code    | MAT5CJ303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                           |                    |             |
| Course Title   | <b>COMPLEX ANALYSIS I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                           |                    |             |
| Type of Course | <b>Major</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                           |                    |             |
| Semester       | V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                           |                    |             |
| Academic Level | 300-399                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |                    |             |
| Course Details | Credit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Lecture/Tutorial per week | Practicum per week | Total Hours |
|                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4                         | -                  | 60          |
| Pre-requisites | Basics of Real Number System and Calculus.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                           |                    |             |
| Course Summary | This course begins with the concepts of complex numbers. complex plane, polar form of complex numbers, powers and roots, etc. Next we discuss complex functions including power functions and nth root functions. Then we discuss limits, continuity, differentiability and analyticity of complex functions. Cauchy Riemann equations and Harmonic conjugates are also studied. Finally the course discusses some standard complex functions like Exponential functions, Logarithmic functions, Trigonometric and Hyperbolic functions. |                           |                    |             |

**Course Outcomes (CO):**

| CO                                                                                                                                                                                             | CO Statement                                                                                                                                    | Cognitive Level* | Knowledge Category# | Evaluation Tools used                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|------------------------------------------------------------|
| CO1                                                                                                                                                                                            | Understand and explain the properties and representations of complex numbers, including their polar form and operations.                        | U                | C                   | Internal Exam/<br>Assignment/Seminar/<br>Viva/End Sem Exam |
| CO2                                                                                                                                                                                            | Apply the principles of limits, continuity, and differentiability to complex functions and utilize the Cauchy-Riemann equations.                | Ap               | P                   | Internal Exam/<br>Assignment/Seminar/<br>Viva/End Sem Exam |
| CO3                                                                                                                                                                                            | Evaluate and create complex exponential, logarithmic, trigonometric, and hyperbolic functions, understanding their properties and applications. | C                | F                   | Internal Exam/<br>Assignment/Seminar/<br>Viva/End Sem Exam |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M) |                                                                                                                                                 |                  |                     |                                                            |

## Detailed Syllabus:

| Textbook   | <b>Complex Analysis (Third Edition): Dennis G. Zill &amp; Patric D. Shanahan, Jones &amp; Bartlett Learning, 2018.</b> |                                                                                                                                   |           |                           |
|------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------|
| Module     | Unit                                                                                                                   | Content                                                                                                                           | Hrs<br>60 | External<br>Marks<br>(70) |
| <b>I</b>   | <b>Module I</b>                                                                                                        |                                                                                                                                   | <b>13</b> | <b>Min.15</b>             |
|            | 1                                                                                                                      | Section 1.1-Complex Numbers and Their Properties                                                                                  |           |                           |
|            | 2                                                                                                                      | Section 1.2-Complex Plane                                                                                                         |           |                           |
|            | 3                                                                                                                      | Section 1.3- Polar Form of Complex Numbers                                                                                        |           |                           |
|            | 4                                                                                                                      | Section 1.4- Powers and Roots                                                                                                     |           |                           |
|            | 5                                                                                                                      | Section 1.5 -Sets of Points in Complex Plane                                                                                      |           |                           |
| <b>II</b>  | <b>Module II</b>                                                                                                       |                                                                                                                                   | <b>12</b> | <b>Min.15</b>             |
|            | 6                                                                                                                      | Section 2.1 -Complex Functions                                                                                                    |           |                           |
|            | 7                                                                                                                      | Section 2.2- Complex Functions as Mappings- up to and including Example 4.                                                        |           |                           |
|            | 8                                                                                                                      | Section 2.4- Special Power Functions- The Power Function $z^n$ (All the topics in 2.4.1)                                          |           |                           |
|            | 9                                                                                                                      | Section 2.4- Special Power Functions-The power function $z^{\frac{1}{n}}$ (Topics in 2.4.2, up to and including Example 5.)       |           |                           |
|            | 10                                                                                                                     | Section 2.4- Special Power Functions-Principal nth Root Functions and Example 9.                                                  |           |                           |
| <b>III</b> | <b>Module III</b>                                                                                                      |                                                                                                                                   | <b>15</b> | <b>Min.20</b>             |
|            | 11                                                                                                                     | Section 3.1- Limits and Continuity-Limits (All the topics in 3.1.1)                                                               |           |                           |
|            | 12                                                                                                                     | Section 3.1- Limits and Continuity-Continuity (Topics in 3.1.2, up to Example 7. )                                                |           |                           |
|            | 13                                                                                                                     | Section 3.1-Limits and Continuity-Continuity (Theorem 3.1.4 to up to and including a bounding property.                           |           |                           |
|            | 14                                                                                                                     | Section 3.2- Differentiability and Analyticity- up to and including Example 2.                                                    |           |                           |
|            | 15                                                                                                                     | Section 3.2- Differentiability and Analyticity- All the topics after Example 2.                                                   |           |                           |
|            | 16                                                                                                                     | Section 3.3- Cauchy-Riemann Equations-up to and including Theorem 3.3.2                                                           |           |                           |
|            | 17                                                                                                                     | Section 3.3 - Cauchy Riemann Equations: -All the topics after Theorem 3.3.2.                                                      |           |                           |
|            | 18                                                                                                                     | Section 3.4 - Harmonic Functions                                                                                                  |           |                           |
| <b>IV</b>  | <b>Module IV</b>                                                                                                       |                                                                                                                                   | <b>8</b>  | <b>Min.15</b>             |
|            | 19                                                                                                                     | Section 4.1 Exponential and Logarithmic Functions- Complex Exponential Function (Topics in 4.1.1 up to and including Periodicity) |           |                           |

|   |                       |                                                                                                                                                  |    |  |
|---|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----|--|
|   | 20                    | Section 4.1 Exponential and Logarithmic Functions-Complex Logarithmic Function (Topics in 4.1.2 up to and including Example 4)                   |    |  |
|   | 21                    | Section 4.3 Trigonometric and Hyperbolic Functions-Complex Trigonometric Functions (Topics in 4.3.1, up to and excluding trigonometric mapping.) |    |  |
|   | 22                    | Section 4.3 Trigonometric and Hyperbolic Functions-Complex Hyperbolic Functions (All the topics in 4.3.2)                                        |    |  |
| V | Module V (Open Ended) |                                                                                                                                                  | 12 |  |
|   |                       | Linear Mappings, Reciprocal Functions                                                                                                            |    |  |
|   |                       | Branches, Branch Cuts and Points, Complex Powers                                                                                                 |    |  |
|   |                       | Inverse Trigonometric and Hyperbolic Functions.                                                                                                  |    |  |

References

1. Brown, James Ward, and Ruel V. Churchill. Complex variables and applications. McGraw-Hill, 2009.

2. Stein, Elias M., and Rami Shakarchi. Complex analysis. Vol. 2. Princeton University Press, 2010.

3. Burckel, Robert B. An Introduction to Classical Complex Analysis: Vol. 1. Vol. 64. Birkhäuser, 2012

4. Hormander, Lars. An introduction to complex analysis in several variables. Elsevier, 1973.

5. Priestley, Hilary A. Introduction to complex analysis. OUP Oxford, 2003.

6. Silverman, Richard A. Introductory complex analysis. Courier Corporation, 2013

7. Bak, Joseph, Donald J. Newman, and Donald J. Newman. *Complex analysis*. Vol. 8. New York: Springer, 2010.

**\*70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

**Mapping of COs with PSOs and POs :**

|      | PSO1 | PSO2 | PSO3 | PSO4 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|------|------|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| CO 1 | 3    | 1    | 0    | 0    | 3   | 0   | 0   | 0   | 2   | 0   | 0   |
| CO 2 | 0    | 3    | 1    | 0    | 2   | 0   | 3   | 0   | 3   | 0   | 0   |
| CO 3 | 1    | 0    | 3    | 0    | 2   | 0   | 3   | 0   | 3   | 0   | 0   |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| -     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

**Mapping of COs to Assessment Rubrics:**

|      | Internal Exam | Assignment | Seminar | Viva | End Semester Examinations |
|------|---------------|------------|---------|------|---------------------------|
| CO 1 | ✓             | ✓          | ✓       | ✓    | ✓                         |
| CO 2 | ✓             | ✓          | ✓       | ✓    | ✓                         |
| CO 3 | ✓             | ✓          | ✓       | ✓    | ✓                         |